

TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature, on *which* lead, a clinician can trust. We give a *target-specific recoverability map*: for a waveform segment s (P/QRS/ST/T), an observed lead set S , and a target lead ℓ , two closed-form numbers derived from one truncated SVD of the segment’s empirical rank-3 spatial (“dipolar”) sub-matrix—an *identifiability* $\eta_{s,\ell}(S)$ (zero iff the target’s low-rank component is recoverable from S) and a *conditioning* $\kappa_{s,\ell}(S)$ (its noise/observed-residual gain). The per-lead criterion is the classical estimability condition specialized to ECG, but made *graded*: on PTB-XL the map ranks target leads on a continuum by *normalized* identifiability and expected mV ambiguity—e.g. from limb leads, anterior precordial ST (V1–V4) is far from identifiable while lateral V5/V6 are much less ambiguous, though none is *exactly* identifiable—a graded call a physiological precordial/limb split cannot make. It is stable to the truncation tolerance for well-posed configurations and rank-sensitive (with reported sensitivity) for ill-posed ones. We attach distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression, strict fold discipline, honestly stated within-group coverage), and observe that the total ST-threshold error rate is empirically similar across the evaluated reconstructors (not a certified bound) while the false-positive/false-negative split is a reconstructor choice. The map turns “trust the reconstruction” into a per-feature, per-lead, calibrated statement.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for*

a given observed lead set, which target lead’s dipolar morphology is even identifiable, and how well-conditioned is its recovery? Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis M_s (top-3 singular vectors of the population lead covariance). We stress that M_s is a *population-estimated (PCA)* object: on real PTB-XL it shares two directions with the classical vectorcardiographic (inverse-Dower) dipole but its third direction differs materially, so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 4).

Related work and contributions. Classical work reconstructs missing leads (regression / VCG synthesis [3, 4, 5], deep and arbitrary-mask waveform-completion models [1, 2, 6, 7]) or *selects* optimal leads [8]; whether a given target is even recoverable is the classical estimability question [9]. Those systems produce a reconstruction; we instead certify, before any reconstruction, which target is recoverable and attach a graded, per-target-lead recoverability measure with calibrated per-record error. Our neural baseline is a *representative* masked-completion U-Net (same family as [1, 6]), not a state-of-the-art claim. We do not claim the underlying identity as new. Our contributions are: (1) a *graded*, per-target-lead recoverability map—continuous $\eta_{s,\ell}(S)/\kappa_{s,\ell}(S)$ from a single truncated SVD that *rank* target leads (not a binary rank test), over an *empirically-estimated* M_s that differs 43–55° from the textbook inverse-Dower dipole, with a truncation-tolerance sensitivity analysis and record-bootstrap CIs (Sec. 2); (2) a calibration layer attaching distribution-free intervals (a sound application of conformalized quantile regression + Mondrian [10], not a new method) to the map per feature and lead, with strict fold discipline and honest within-group (not per-diagnosis) coverage (Sec. 3); (3) a graded ST-safety study on PTB-XL—continuous full-waveform reconstruction shows anterior precordial ST is far from recoverable from limb leads, with a total ST-threshold error rate empirically similar across the evaluated reconstructors (an empirical observation, not a certified bound) and a false-positive/false-negative split that is a reconstructor choice (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Model. Within segment s write the population 12-lead vector as $\mathbf{L}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{d} + \mathbf{r}_s$, where $\mathbf{M}_s \in \mathbb{R}^{12 \times 3}$ is the (population-estimated, PCA) dipolar basis, $\mathbf{d}$ the dipole coordinates, and $\mathbf{r}_s$ the non-dipolar residual. Observing S gives $\mathbf{y}_S = \text{Sel}_S \mathbf{L}_s + \mathbf{n}$. Let $\mathbf{M}_{s,S} = \text{Sel}_S \mathbf{M}_s$ have truncated SVD at relative tolerance ϱ ; $\mathbf{M}_{s,S}^+$ its truncated pseudo-inverse and $\mathbf{P}_{\text{obs}} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S}$ the projector onto the observed dipole coordinate directions.

Proposition 1 (Per-target-lead dipolar identifiability and conditioning). *The mean-centred dipolar estimate $\hat{\mathbf{L}}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{M}_{s,S}^+ (\mathbf{y}_S - \boldsymbol{\mu}_{s,S})$ satisfies, for each target lead ℓ with $\mathbf{e}_\ell$ the ℓ -th unit vector:*

- (i) *Identifiability. Define $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$. If $\eta_{s,\ell}(S) = 0$, the dipolar component of lead ℓ is identifiable from S : $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ is determined by $\text{Sel}_S \mathbf{M}_s \mathbf{d}$. If $\eta_{s,\ell}(S) > 0$, there is a dipole direction unobserved by S that changes lead ℓ ; its dipolar component is not identifiable at any SNR.*
- (ii) *Conditioning. On the identifiable part, the error from observation noise and from the observed non-dipolar residual is amplified into lead ℓ by $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$: $|\mathbf{e}_\ell^\top (\hat{\mathbf{L}}_s - \boldsymbol{\mu}_s - \mathbf{M}_s \mathbf{P}_{\text{obs}} \mathbf{d})| \leq \kappa_{s,\ell}(S) (\|\text{Sel}_S \mathbf{r}_s\| + \|\mathbf{n}\|)$.*

The global $\kappa_s(S) = \|\mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$ is a configuration-level (spectral) worst case: since a row norm never exceeds the spectral norm, $\kappa_{s,\ell}(S) \leq \kappa_s(S)$ for every ℓ , so κ_s upper-bounds every per-lead conditioning (it is not their maximum). Both $\eta_{s,\ell}$ and $\kappa_{s,\ell}$ depend on the estimated $\mathbf{M}_s$ and on ϱ ; near-rank-deficient S are reported with an ϱ -sensitivity sweep and record-bootstrap confidence intervals.

Relation to classical estimability. Part (i) is the classical estimability criterion (Gauss–Markov / [9]) specialized to the per-lead linear functional $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$: $\eta_{s,\ell}(S) = 0$ iff that functional lies in the row space of $\mathbf{M}_{s,S}$, and $\kappa_{s,\ell}$ is its coefficient norm. We claim no novelty for this identity. The contribution (Sec. 4) is its use as a *graded*, per-target-lead, *empirically-estimated* recoverability map: a continuous η that ranks target leads (not a binary rank test), a subspace $\mathbf{M}_s$ that differs materially from the textbook inverse-Dower dipole, and per-record calibrated error rates (Sec. 3) attached to it—quantities not read off lead geometry.

The non-dipolar residual (no independence claim on real data). Let $\mathbf{m}_s(\mathbf{y}_S) = \mathbb{E}[\mathbf{r}_s | \mathbf{y}_S]$ and $\mathbf{u}_s = \mathbf{r}_s - \mathbf{m}_s(\mathbf{y}_S)$, so $\mathbb{E}[\mathbf{u}_s | \mathbf{y}_S] = \mathbf{0}$. We do *not* assume $\mathbf{u}_s \perp \mathbf{y}_S$. The Bayes-optimal reconstruction error of the residual is

$$\inf_{\hat{\mathbf{r}}} \mathbb{E} \|\mathbf{r}_s - \hat{\mathbf{r}}(\mathbf{y}_S)\|^2 = \mathbb{E} [\text{Var}(\mathbf{r}_s | \mathbf{y}_S)].$$

Part of $\mathbf{r}_s$ (the *empirically predictable residual*) is recovered by a predictor trained on $\mathbf{y}_S$ and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—not declared unrecoverable a priori.

Proposition 2 (Strict independence limit – synthetic model only). *If the data are generated so that a component $\mathbf{u}$ is statistically independent of $\mathbf{y}_S$ (an explicit synthetic construction), then $p(\mathbf{u} | \mathbf{y}_S) = p(\mathbf{u})$, the Bayes estimate of $\mathbf{u}$ is its prior mean, and $\mathbb{E} \|\hat{\mathbf{u}} - \mathbf{u}\|^2 \geq \text{Var}(\mathbf{u})$ for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

3. DISTRIBUTION-FREE CALIBRATION

3.1. Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment’s samples). We train a gradient-boosted quantile regressor of it on the observed leads’ segment means and apply Mondrian conformalized quantile regression [10] per (S, s, ℓ) group, with strict fold discipline: basis and model on folds 1–7, the regressor’s `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal $1-\alpha=0.90$). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered (S, s, ℓ) groups, fold-10 within-group coverage clusters near nominal (0.88–0.95; e.g. {I,II,V1,V3,V5} QRS/V2 0.91, record-bootstrap CI [0.89, 0.93], mean width 0.150 mV).

What is and is not guaranteed. Mondrian CQR gives *within*-(S, s, ℓ)-group marginal coverage under exchangeability—not coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 ($n=85$), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

4. EXPERIMENTS

Data. PTB-XL [11] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [12].

Reproducibility. Code, fixed and deterministic seeds, an exact environment lock, result lineage (commit / dataset / split hashes in every JSON), and a one-command pipeline are re-

leased¹; every number in the tables is regenerated from result JSON (no hand-typed values). The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual Group-Norm blocks, 60 epochs, Adam 2×10^{-4} , MSE on masked leads under random masks, 3 training seeds); the residual predictor is a gradient-boosted quantile regressor. The map is a deterministic function of the estimated μ_s, M_s (record-level bootstrap CIs), not a held-out evaluation.

4.1. The recoverability map on PTB-XL

We estimate M_s (rank 3) per segment on the training NORM records and report the map with RECORD-LEVEL bootstrap CIs (each resample draws whole records, re-pools their segment samples, and refits M_s ; truncation tolerance $\varrho=10^{-2}$; Fig. 1). The map is a deterministic function of the estimated M_s and the selection S ; the CI quantifies its estimation uncertainty. A single observed lead leaves every other lead’s dipolar component unidentifiable; a dipole-spanning triplet {I,II,V2} or the five-lead {I,II,V1,V3,V5} makes all targets *exactly* identifiable ($\eta=0$ to machine precision, since S spans the rank-3 subspace).

Exact vs. graded identifiability. Exact identifiability is the binary test $\eta_{s,\ell}(S)=0$. For limb-6, *no* precordial lead is exactly identifiable ($\eta_{ST,\ell}>0$ for all), so we grade the shortfall with two scale-aware quantities rather than raw η : the *normalized* identifiability $\tilde{\eta}_{s,\ell} = \eta_{s,\ell} / \|e_\ell^\top M_s\|_2 \in [0, 1]$ (the fraction of the lead’s dipolar content that is unobservable) and the *prior-conditional expected ambiguity* in mV—the residual a Bayes reconstructor still incurs on the dipolar component after conditioning the unobserved dipole coordinate on the observed one through the fitted (Gaussian) dipole prior, $\sqrt{e_\ell^\top M_s \Sigma_{Q|P} M_s^\top e_\ell}$ with $\Sigma_{Q|P}$ the posterior covariance of the unobserved coordinate. Both fall from strongly unrecoverable anterior leads (V2 $\tilde{\eta}=0.960$, ambiguity 0.200 mV; V3, V1, V4) to less-ambiguous lateral leads (V5 $\tilde{\eta}=0.170$, 0.024 mV; V6 $\tilde{\eta}=0.057$, 0.007 mV). We therefore say **anterior precordial ST (V1–V4) is far from identifiable from limb leads, while lateral V5/V6 are much less ambiguous—but none is exactly identifiable**. A physiological precordial/limb split cannot make this graded call; the empirical map does.

Downstream ST cost across reconstructors. We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads kept exact; ST measured at J+60 ms vs. a PR baseline on fiducials located once on the observed Lead II and shared between truth and reconstruction), and count ST-threshold events ($|\text{ST}| \geq 0.1$ mV, elevation or depression; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 1). The *total* wrong-event rate is *empirically similar* across the evaluated reconstructors (45.8–47.3%); all three *miss* more true

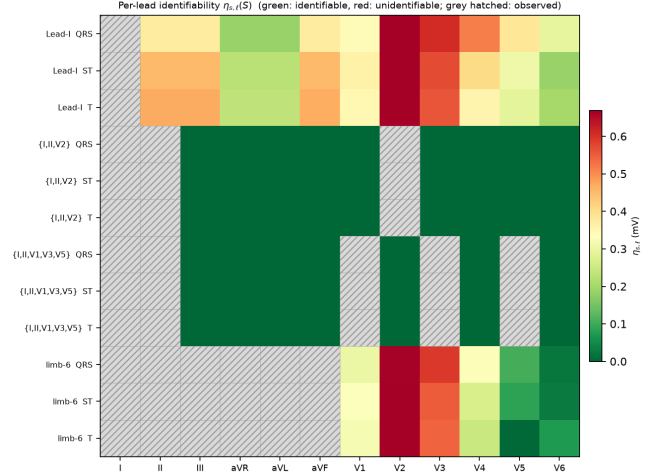


Fig. 1. Per-lead identifiability $\eta_{s,\ell}(S)$ across configurations and segments (green: identifiable, red: unidentifiable; grey hatched: observed). For limb-6 the anterior precordial leads are the most unidentifiable; lateral V5/V6 are less ambiguous.

Table 1. Limb-6 → anterior-precordial ST reconstruction, 1498 fold-10 records. FP = false-positive, FN = false-negative ST-threshold crossings (%); total = FP+FN. The total is empirically similar across reconstructors; the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

reconstructor	FP %	FN %	total %	ST err (mV)
dipolar	14.5	32.8	47.3	0.076
ridge	6.9	38.9	45.8	0.072
OLS	6.2	39.6	45.9	0.073

crossings than they invent (FN>FP)—limb→precordial reconstruction smooths ST toward the population and drops real deviations—and the FP/FN balance shifts with the reconstructor (the dipolar recovery has the largest false-positive share, OLS the smallest). We report the whole matrix; because we do not derive a minimax/Bayes lower bound, we describe the shared total as an empirically similar total error rate across these three reconstructors—not a certified floor.

Truncation-tolerance sensitivity. Rank/ κ of well-posed configurations are stable across $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$ (e.g. {I,II,V1,V3,V5} is rank 3 throughout), while near-rank-deficient sets are sensitive: on synthetic data {V1,V2,V3} is rank 2 at $\varrho=10^{-2}$ but rank 3 with $\kappa \sim 200$ at $\varrho=10^{-4}$. We therefore report a single κ only for well-conditioned configurations and give the ϱ -sweep and bootstrap CIs otherwise; a single 10^4 or 10^5 figure for a degenerate set is not meaningful.

¹ Anonymized for review; released on acceptance.

Table 2. Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

config	prior-mean	dipolar	ridge	U-Net
{I,II,V2}	0.467	0.236	0.220	0.183
{I,II,V1,V3,V5}	0.481	0.183	0.167	0.164
limb-6	0.481	0.361	0.344	0.238

4.2. Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge λ selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead masks (one model serves any configuration, no target-lead leakage), the same family as [1, 6, 7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge→U-Net step is within its paired CI)—prior-mean → dipolar (Tier I) → per-segment ridge (Tier I+II) → U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on {I,II,V1,V3,V5}; paired Δ CI $[-0.007, +0.000]$ mV). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS 0.481 → 0.238)—it recovers the predictable, largely population-correlated structure—but plateaus at $1.46\times$ its spanning-set error (paired spanning→limb-6 Δ CI $[+0.070, +0.079]$ mV) and no reconstructor closes that gap. This residual gap is *consistent with* the unobserved dipolar coordinate the map’s $\eta > 0$ marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equality): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket “not recoverable.”

Empirical, not physical, subspace. On real PTB-XL the estimated M_s shares two directions with the classical inverse-Dower vectorcardiographic space [3, 4] (principal angles 2–6°) but its third direction differs materially (max angle 43–55°, bootstrap CIs excluding small angles). We therefore treat M_s as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quan-

ties are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

5. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a graded per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It tells a clinician which lead configuration can support a decision, e.g. that anterior precordial ST is far from readable from limb leads while lateral V5/V6 are much less ambiguous, and reports the empirically similar total ST-threshold error rate the evaluated reconstructors incur there.

6. REFERENCES

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